

Modelado genómico del metabolismo bacteriano

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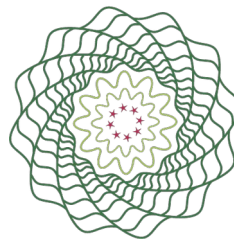
3a Escuela Queretana de Matemáticas

Taller de Biomatemáticas

18 de junio de 2026



LIIGH UNAM



Escuela
Queretana de
Matemáticas

¿Cómo crecen las poblaciones de células?

Plan

- Las primeras mediciones y el modelo de Monod
- Un modelo a partir de principios básicos
- El Análisis de Balance de Flujos (FBA)
- Conocimiento biológico: la matriz estequiométrica
- Incorporando interacciones, el Balance de Flujos dinámico (DFBA)
- Descanso (~30 min)
- Demostración
- Ejercicio

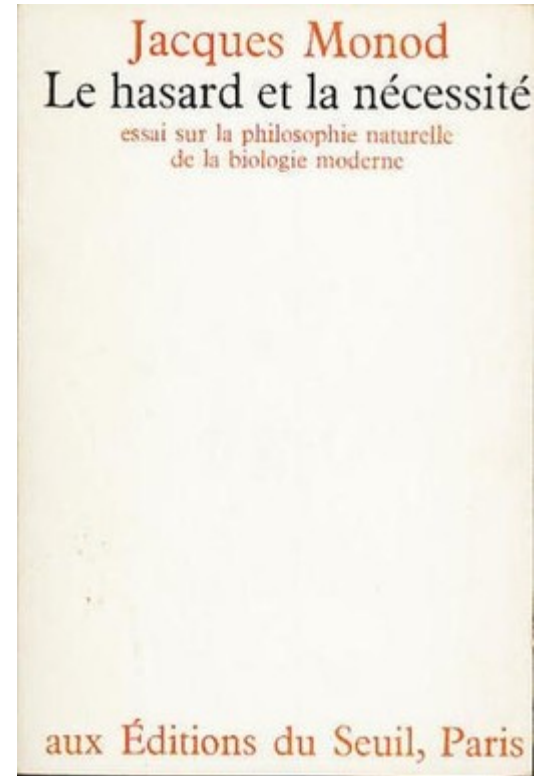
Jacques Monod

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The Nobel Prize in Physiology or Medicine 1965 was awarded jointly to François Jacob, André Lwoff and Jacques Monod "for their discoveries concerning genetic control of enzyme and virus synthesis"

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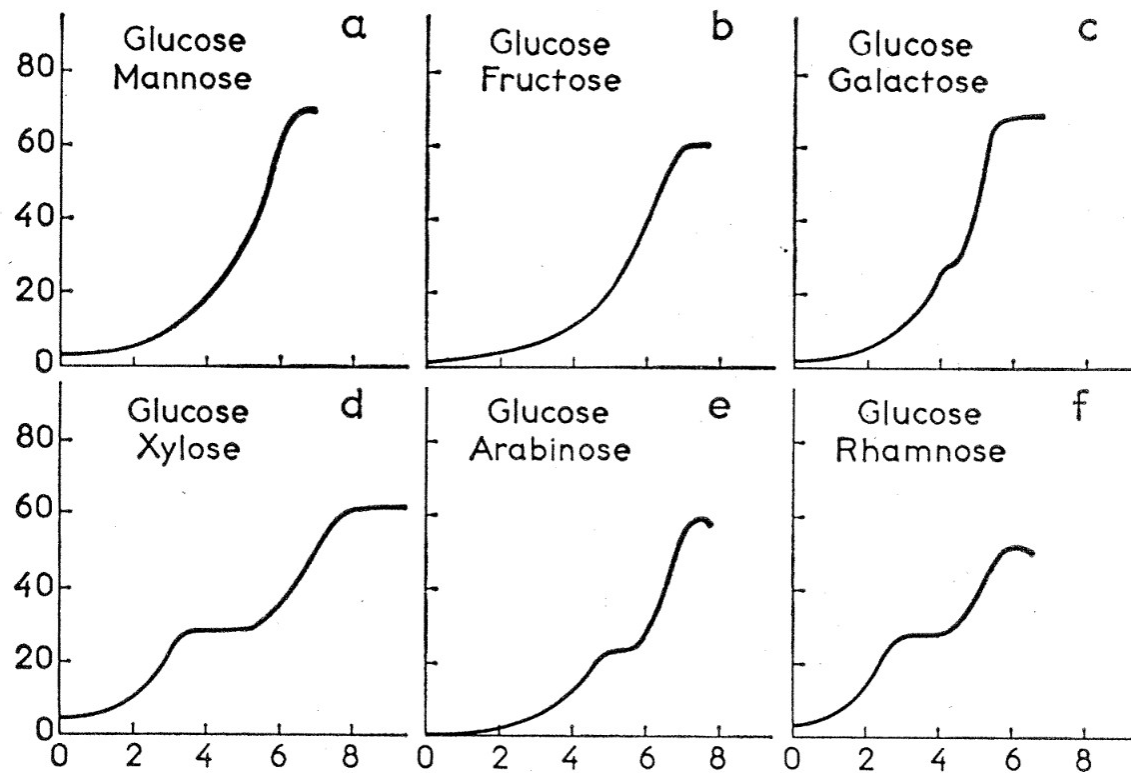


Fig. 1. Growth of *Escherichia coli* in the presence of different carbohydrate pairs serving as the only source of carbon in a synthetic medium (50).

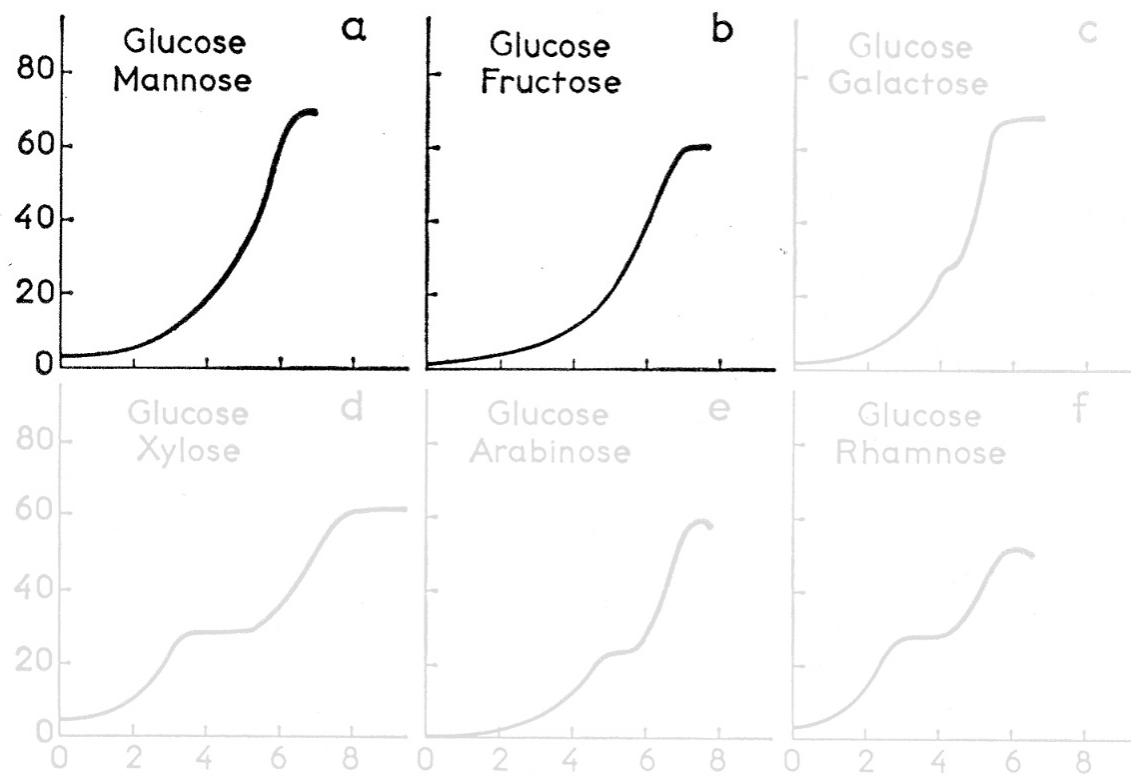


Fig. 1. Growth of *Escherichia coli* in the presence of different carbohydrate pairs serving as the only source of carbon in a synthetic medium (50).

La ecuación de Monod

$$\frac{\partial X}{\partial t} = \mu = \mu_{max} \frac{[S]}{K_s + [S]}$$

μ is the growth rate of a considered microorganism,

μ_{max} is the maximum growth rate of this microorganism,

$[S]$ is the concentration of the limiting substrate S for growth,

K_s is the “half-velocity constant”—the value of $[S]$ when $\mu/\mu_{max} = 0.5$.

Análisis de balance de flujos (FBA)

$$\max_{\mathbf{v}} \mu = \mathbf{w}^T \mathbf{v}$$

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

$$\mathbf{v}_{\min} \leq \mathbf{v} \leq \mathbf{v}_{\max}$$

$\mathbf{A}$ = matriz estequiométrica

$\mathbf{v}$ = flujos de cada reacción

μ = tasa de crecimiento

$\mathbf{w}$ = contribución de cada reactivo al crecimiento

$\mathbf{v}_{\min}$, $\mathbf{v}_{\max}$ = límites biológicos a los flujos

Análisis de balance de flujos dinámico (DFBA)

$$\mathbf{v}_s = \mathbf{f}_s(\mathbf{S}, \mathbf{P})$$



$$\max_v \mu = \mathbf{w}^T \mathbf{v}$$

$$\mathbf{A} \mathbf{v} = \mathbf{0}$$

$$\mathbf{v}_{\min} \leq \mathbf{v} \leq \mathbf{v}_{\max}$$

$$\mathbf{v}_p$$



$$\frac{dX}{dt} = \mu X$$

$$\frac{d\mathbf{S}}{dt} = -\mathbf{v}_s X$$

$$\frac{d\mathbf{P}}{dt} = \mathbf{v}_p X$$

Análisis de balance de flujos dinámico (DFBA) para comunidades

$$\mathbf{v}_s = \sum_i \mathbf{v}_{s,i} = \sum_i \mathbf{f}_{s,i}(\mathbf{S}, \mathbf{P})$$



$$\begin{aligned} \max_v \mu &= \sum \mu_i = \mathbf{w}^T \mathbf{v} \\ \mathbf{A} \mathbf{v} &= \mathbf{0} \\ \mathbf{v}_{\min} &\leq \mathbf{v} \leq \mathbf{v}_{\max} \end{aligned}$$



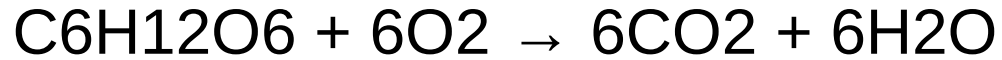
$$\mathbf{v}_{p,i}$$

$$\frac{dX_i}{dt} = \mu_i X_i$$

$$\frac{d\mathbf{S}}{dt} = - \sum_i \mathbf{v}_{s,i} X_i$$

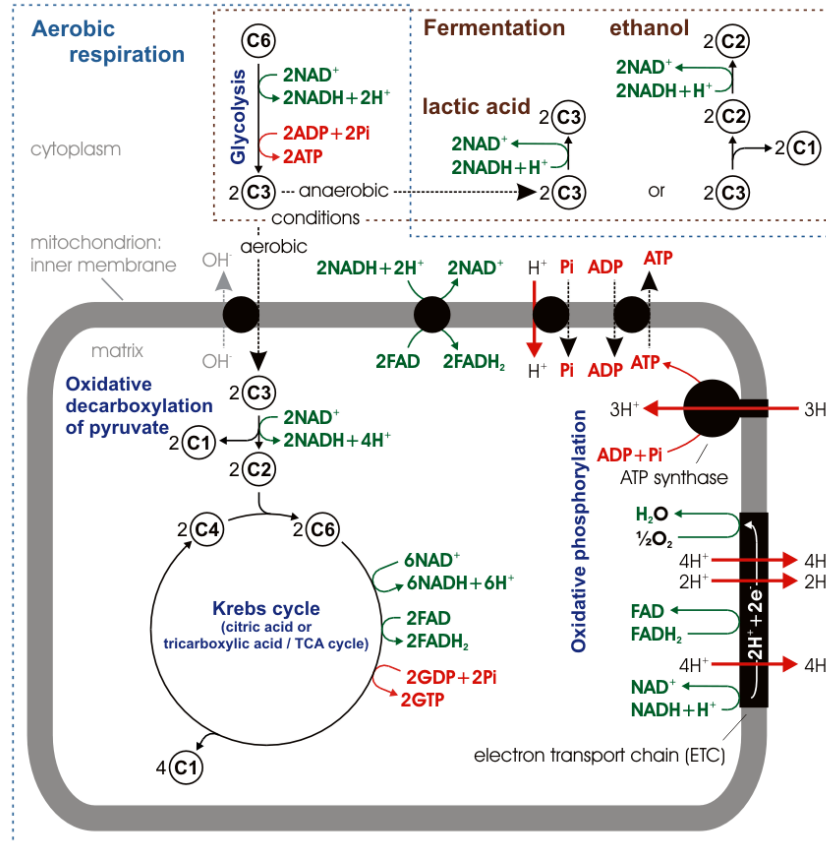
$$\frac{d\mathbf{P}}{dt} = \sum_i \mathbf{v}_{p,i} X_i$$

Reacciones como vectores estequiométricos



$$\mathbf{A} = \begin{bmatrix} -1 \\ -6 \\ 6 \\ 6 \end{bmatrix}$$

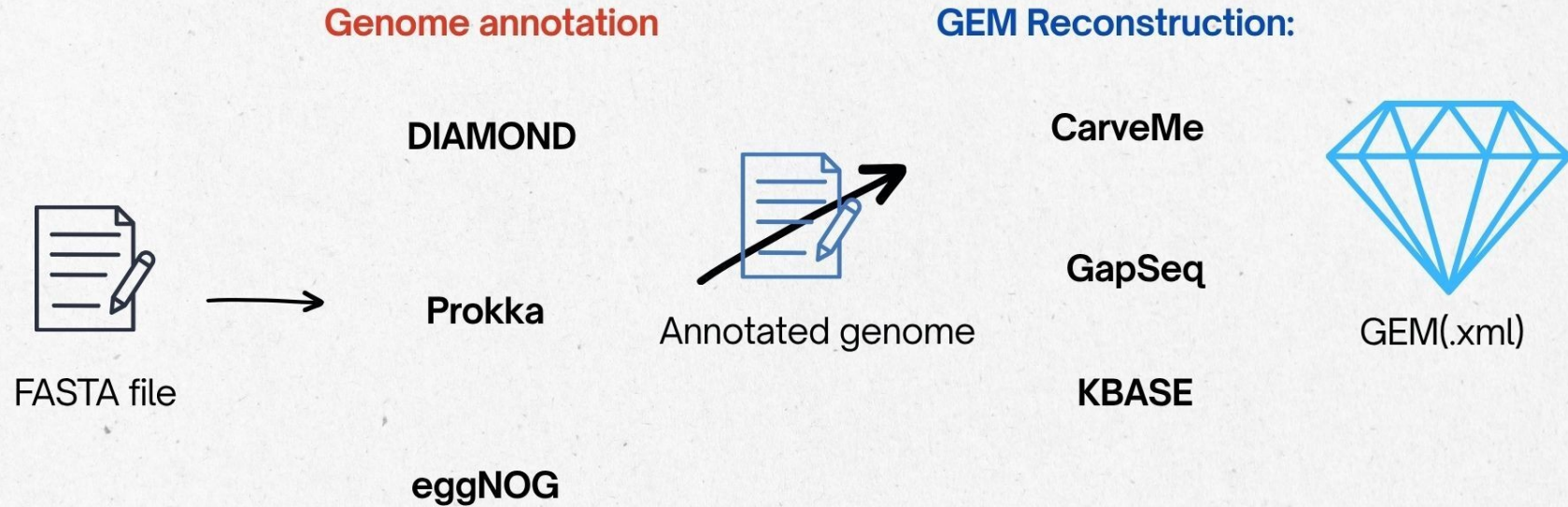
Complejidad biológica



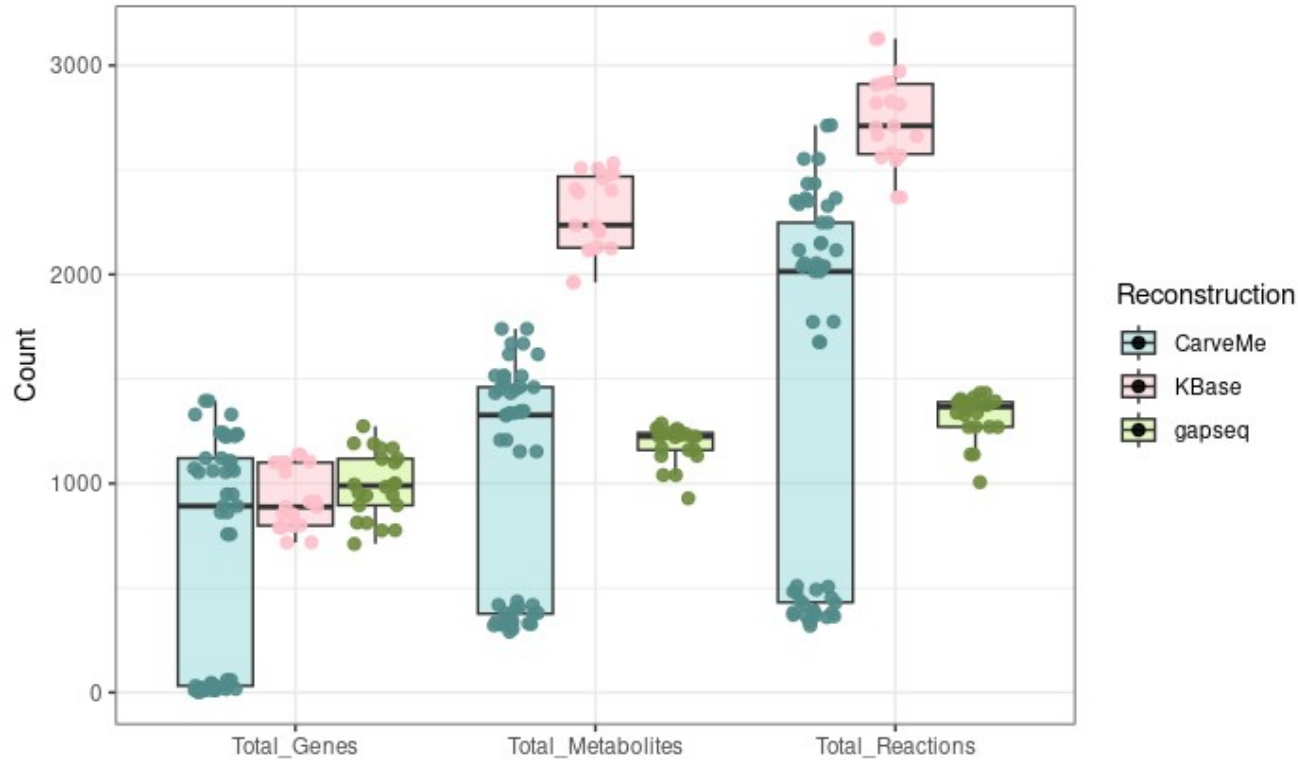
Complejidad biológica

	Glycolytic reactions											AMP metabolism		Primary export		Co-factors		Primary inputs		Inorganic		
	v_{HK}	v_{PGI}	v_{PFK}	v_{TPI}	v_{ALD}	v_{GAPDH}	v_{PGK}	v_{PGLM}	v_{ENO}	v_{PK}	v_{LDH}	v_{AMP}	v_{APK}	v_{PYR}	v_{LAC}	v_{ATP}	v_{NADH}	v_{GLUin}	v_{AMPin}	v_{H^+}	v_{H_2O}	ρ_i
Glu	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	2
G6P	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
F6P	0	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
FBP	0	0	1	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
DHAP	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
GAP	0	0	0	1	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3
PG13	0	0	0	0	0	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
PG3	0	0	0	0	0	0	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	2
PG2	0	0	0	0	0	0	0	1	-1	0	0	0	0	0	0	0	0	0	0	0	0	2
PEP	0	0	0	0	0	0	0	0	1	-1	0	0	0	0	0	0	0	0	0	0	0	2
PYR	0	0	0	0	0	0	0	0	0	1	-1	0	0	-1	0	0	0	0	0	0	0	3
LAC	0	0	0	0	0	0	0	0	0	0	1	0	0	0	-1	0	0	0	0	0	0	2
NAD	0	0	0	0	0	-1	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	3
NADH	0	0	0	0	0	1	0	0	0	0	-1	0	0	0	0	0	-1	0	0	0	0	3
AMP	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	1	0	0	3
ADP	1	0	1	0	0	0	-1	0	0	-1	0	0	-2	0	0	1	0	0	0	0	0	6
ATP	-1	0	-1	0	0	0	1	0	0	1	0	0	1	0	0	-1	0	0	0	0	0	6
P_i	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	2
H^+	1	0	1	0	0	1	0	0	0	-1	-1	0	0	0	0	1	1	0	0	-1	0	8
H_2O	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0	0	0	-1	3
π_j	5	2	5	2	3	5	4	2	3	5	5	1	3	1	1	5	3	1	1	1	1	
C	0	0	0	0	0	0	0	0	0	0	0	-10	0	-3	-3	0	0	6	10	0	0	
H	0	0	0	0	0	0	0	0	0	0	0	-13	0	-3	-5	0	0	12	13	-1	-2	
O	0	0	0	0	0	0	0	0	0	0	0	-7	0	-3	-3	0	0	6	7	0	-1	
P	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	1	0	0	
N	0	0	0	0	0	0	0	0	0	0	0	-5	0	0	0	0	0	0	5	0	0	
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
NAD	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

GEM RECONSTRUCTION METHOD AND ANNOTATION APPROACH



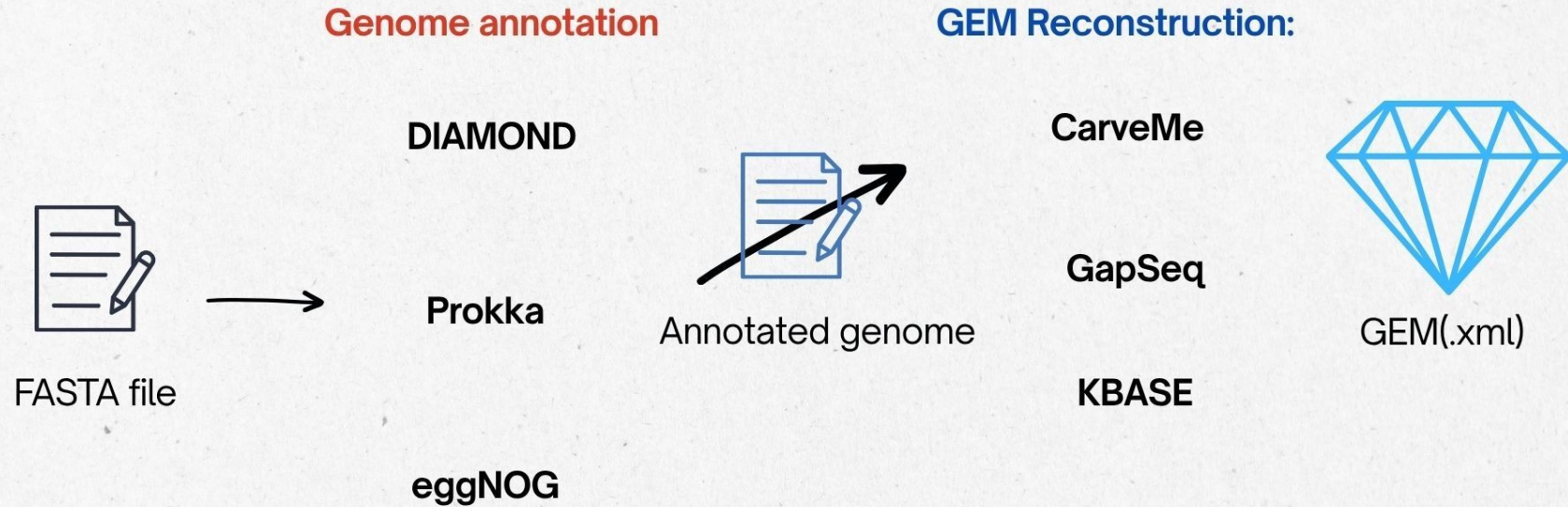
Ningún método automatizado es suficientemente bueno



Bases de datos

- BiGG
- ModelSEED
- KBase
- AGORA2
- COBRA
- ...

GEM RECONSTRUCTION METHOD AND ANNOTATION APPROACH



Reacciones en los GEMs

```
<reaction metaid="meta_R_FLVR" sboTerm="SBO:0000176" id="R_FLVR" name="Flavin reductase" reversible="false" fast="false" fbc:lowerFluxBound="cobra_0_bound" fbc:upperFluxBound="cobra_default_ub">
  <annotation>
    <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:vCard="http://www.w3.org/2001/vcard-rdf/3.0#" xmlns:vCard4="http://www.w3.org/2006/vcard/ns#" xmlns:bqbiol="http://biomodels.net/biology-qualifiers/" xmlns:bqmodel="http://biomodels.net/model-qualifiers/"
      <rdf:Description rdf:about="#meta_R_FLVR">
        <bqbiol:is>
          <rdf:Bag>
            <rdf:li rdf:resource="https://identifiers.org/ec-code/1.5.1.30"/>
            <rdf:li rdf:resource="https://identifiers.org/ec-code/1.5.1.41"/>
            <rdf:li rdf:resource="https://identifiers.org/rhea/19377"/>
            <rdf:li rdf:resource="https://identifiers.org/rhea/19378"/>
            <rdf:li rdf:resource="https://identifiers.org/rhea/19379"/>
            <rdf:li rdf:resource="https://identifiers.org/rhea/19380"/>
            <rdf:li rdf:resource="https://identifiers.org/biocyc/META:NADPH-DEHYDROGENASE-FLAVIN-RXN"/>
            <rdf:li rdf:resource="https://identifiers.org/metanetx.reaction/MNXR99601"/>
            <rdf:li rdf:resource="https://identifiers.org/kegg.reaction/R05707"/>
            <rdf:li rdf:resource="https://identifiers.org/seed.reaction/rxn03976"/>
          </rdf:Bag>
        </bqbiol:is>
      </rdf:Description>
    </rdf:RDF>
  </annotation>
  <listOfReactants>
    <speciesReference species="M_h_c" stoichiometry="1" constant="true"/>
    <speciesReference species="M_nadph_c" stoichiometry="1" constant="true"/>
    <speciesReference species="M_ribflv_c" stoichiometry="1" constant="true"/>
  </listOfReactants>
  <listOfProducts>
    <speciesReference species="M_nadp_c" stoichiometry="1" constant="true"/>
    <speciesReference species="M_rbflvrd_c" stoichiometry="1" constant="true"/>
  </listOfProducts>
  <fbc:geneProductAssociation>
    <fbc:or>
      <fbc:geneProductRef fbc:geneProduct="G_NP_418286_1"/>
    </fbc:or>
    <fbc:and>
      <fbc:geneProductRef fbc:geneProduct="G_NP_417243_1"/>
      <fbc:geneProductRef fbc:geneProduct="G_NP_417244_1"/>
    </fbc:and>
  </fbc:or>
</fbc:geneProductAssociation>
</reaction>
```

Pesos en los GEMs

```
</reaction>
<reaction metaid="meta_Growth" sboTerm="SBO:0000629" id="Growth" name="Biomass reaction" reversible=
  <listOfReactants>
    <speciesReference species="M_10fthf_c" stoichiometry="0.000223" constant="true"/>
    <speciesReference species="M_ala_L_c" stoichiometry="0.513689" constant="true"/>
    <speciesReference species="M_amet_c" stoichiometry="0.000223" constant="true"/>
    <speciesReference species="M_arg_L_c" stoichiometry="0.295792" constant="true"/>
    <speciesReference species="M_asn_L_c" stoichiometry="0.241055" constant="true"/>
    <speciesReference species="M_asp_L_c" stoichiometry="0.241055" constant="true"/>
    <speciesReference species="M_atp_c" stoichiometry="54.124831" constant="true"/>
    <speciesReference species="M_ca2_c" stoichiometry="0.005205" constant="true"/>
    <speciesReference species="M_cl_c" stoichiometry="0.005205" constant="true"/>
    <speciesReference species="M_coa_c" stoichiometry="0.000576" constant="true"/>
    <speciesReference species="M_cobalt2_c" stoichiometry="0.0001" constant="true"/>
    <speciesReference species="M_ctp_c" stoichiometry="0.133508" constant="true"/>
    <speciesReference species="M_cu2_c" stoichiometry="0.000709" constant="true"/>
    <speciesReference species="M_cys_L_c" stoichiometry="0.09158" constant="true"/>
    <speciesReference species="M_datp_c" stoichiometry="0.026166" constant="true"/>
    <speciesReference species="M_dctp_c" stoichiometry="0.027017" constant="true"/>
    <speciesReference species="M_dgtp_c" stoichiometry="0.027017" constant="true"/>
    <speciesReference species="M_dttp_c" stoichiometry="0.026166" constant="true"/>
    <speciesReference species="M_fad_c" stoichiometry="0.000223" constant="true"/>
    <speciesReference species="M_fe2_c" stoichiometry="0.006715" constant="true"/>
    <speciesReference species="M_fe3_c" stoichiometry="0.007808" constant="true"/>
    <speciesReference species="M_gln_L_c" stoichiometry="0.26316" constant="true"/>
    <speciesReference species="M_glu_L_c" stoichiometry="0.26316" constant="true"/>
    <speciesReference species="M_gly_c" stoichiometry="0.612638" constant="true"/>
    <speciesReference species="M_gtp_c" stoichiometry="0.215096" constant="true"/>
    <speciesReference species="M_h2o_c" stoichiometry="48.601527" constant="true"/>
    <speciesReference species="M_his_L_c" stoichiometry="0.094738" constant="true"/>
    <speciesReference species="M_ile_L_c" stoichiometry="0.290529" constant="true"/>
```

Simulando el crecimiento a partir de GEMs

openCOBRA

Open-source, community-developed code base for CONstraint-Based Reconstruction and Analysis.

Packages



The COBRA Toolbox



COBRApy



COBRA.jl

COMETS

<https://opencobra.github.io/>

<https://www.runcomets.org/home>

Simulando el crecimiento a partir de GEMs

Table 2 | Comparison of COMETS capabilities with previous version and other dynamic FBA based software packages

	COMETS v.1 ²⁸	COMETS v.2 (this work)	BacArena ¹⁰¹	Matnet ¹¹⁴	3DdFBA ¹¹⁵	Spatial dFBA lab ⁴²	IndiMesh ¹¹⁶
Multiple genotypes	✓✓	✓✓	✓✓	✓✓	X	✓	✓✓
Diffusive bacterial/ metabolite spread	✓✓	✓✓	✓✓	✓	✓✓	✓	✓✓
Convective and/or collective bacterial spread	X	✓✓	✓✓	✓	✓✓	✓	✓✓
Chemotaxis	X	X	✓✓	✓	X	✓	✓✓
Spatially varying diffusion	X	✓✓	✓✓	X	X	X	✓✓
Evolution	X	✓✓	X	X	X	X	X
Extracellular enzymes	X	✓✓	X	X	X	✓	X
Approachable toolboxes	X	✓✓	✓✓	✓✓	X	X	X
GUI	✓✓	✓✓	X	X	X	X	X
Population-based or agent- based	Population	Population	Agent	Agent	Population	Population	Agent

A double checkmark indicates a fully functional capability; a single checkmark indicates a limited capability, one that requires additional programming/script writing

COMETSpy básico

```
import cometspy as c
import cobra.io
import matplotlib.pyplot as plt

#Create and set environment (layout)
test_tube = c.layout()
test_tube.set_specific_metabolite('glc__D_e', 0.011)
test_tube.set_specific_metabolite('o2_e', 0)
test_tube.set_specific_metabolite('nh4_e',1000);
test_tube.set_specific_metabolite('pi_e',1000);
test_tube.set_specific_metabolite('h2o_e',1000);
test_tube.set_specific_metabolite('h_e',1000);

# create the model using CobraPy functionality
e_coli_cobra = cobra.io.load_model('textbook')
e_coli = c.model(e_coli_cobra)
e_coli.change_bounds('EX_glc__D_e', -1000, 1000)
e_coli.initial_pop = [0, 0, 5e-6]
test_tube.add_model(e_coli)
```

COMETSpy básico

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import cometspy as c
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#Create and set environment (layout)
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test_tube.set_specific_metabolite('h_e', 1000);

# create the model using CobraPy functionality
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e_coli = c.model(e_coli_cobra)
e_coli.change_bounds('EX_glc__D_e', -1000, 1000)
e_coli.initial_pop = [0, 0, 5e-6]
test_tube.add_model(e_coli)
```

```
# Set the parameters that are different from the default
sim_params = c.params()
sim_params.set_param('defaultVmax', 18.5)
sim_params.set_param('defaultKm', 0.000015)
sim_params.set_param('maxCycles', 1000)
sim_params.set_param('timeStep', 0.01)
sim_params.set_param('spaceWidth', 1)
sim_params.set_param('maxSpaceBiomass', 10)
sim_params.set_param('minSpaceBiomass', 1e-11)
sim_params.set_param('writeMediaLog', True)

# Run experiment
experiment = c.comets(test_tube, sim_params)
experiment.run()
```

Análisis de balance de flujos dinámico (DFBA) para comunidades

$$\mathbf{v}_s = \sum_i \mathbf{v}_{s,i} = \sum_i \mathbf{f}_{s,i}(\mathbf{S}, \mathbf{P})$$



$$\begin{aligned} \max_v \mu &= \sum \mu_i = \mathbf{w}^T \mathbf{v} \\ \mathbf{A} \mathbf{v} &= \mathbf{0} \\ \mathbf{v}_{\min} &\leq \mathbf{v} \leq \mathbf{v}_{\max} \end{aligned}$$

$$\mathbf{v}_{p,i}$$


$$\frac{dX_i}{dt} = \mu_i X_i$$

$$\frac{d\mathbf{S}}{dt} = - \sum_i \mathbf{v}_{s,i} X_i$$

$$\frac{d\mathbf{P}}{dt} = \sum_i \mathbf{v}_{p,i} X_i$$

